

Impact of threshold voltage and gate capacitance on Drain current in MOSFETS

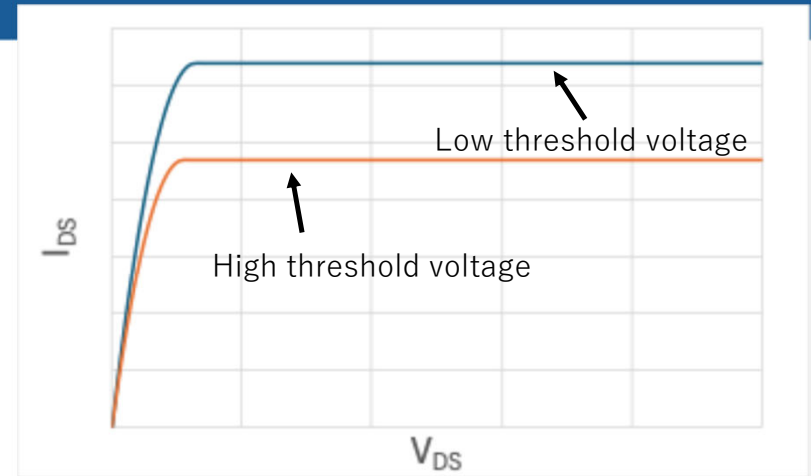
THRESHOLD VOLTAGE

$$I_{DS} = \frac{W}{L} \mu_n C_{ox} \left[(V_{GS} - V_{th}) V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$

$$V_{th} = \phi_{MS} - \frac{Q_s}{C_{ox}} + 2(E_i - E_f) + \frac{\sqrt{4\epsilon_s \epsilon_0 q N_a (E_i - E_f)}}{C_{ox}}$$

$$E_f - E_i = \frac{kT}{q} \log\left(\frac{N_a}{N_i}\right)$$

Q_s : Interface charge
 ϕ_{MS} : Metal-semiconductor work function
 C_{ox} : Oxide capacitance
 E_i : Intrinsic Fermi level
 E_f : Fermi level
 ϵ_s : Interface charge
 ϵ_0 : Vacuum permittivity
 q : Charge of one electron
 N_a : Acceptor doping concentration
 N_i : Intrinsic carrier concentration
 k : Boltzmann constant
 T : Absolute temperature



AS THE THRESHOLD VOLTAGE INCREASES, THE PINCH-OFF VOLTAGE DECREASES, AND THE SATURATION CURRENT ALSO BECOMES SMALLER.

<WHAT FACTORS AFFECT THRESHOLD VOLTAGE?>

COMPATIBILITY BETWEEN METAL AND SEMICONDUCTOR

MATERIAL	WORK FUNCTION(EV)
Al	3.2
Au	4.0
Ti	4.1
W	4.5

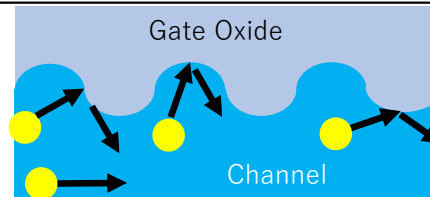
WHY

WHEN THE WORK FUNCTIONS DIFFER, BAND BENDING OCCURS AND ϕ_{MS} VARIES.

RELATED PROCESS

SELECTIONS OF DEPOSITION MATERIAL

SURFACE ROUGHNESS FOR Q_s



WHY

ELECTRONS ARE MORE LIKELY BUMP INTO THE SURFACE AND SCATTER,

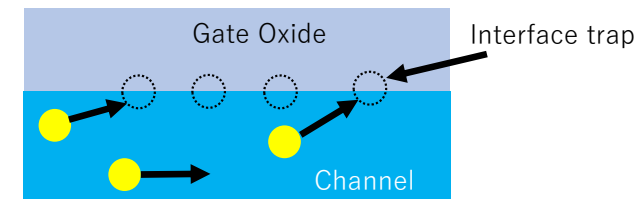
RELATED PROCESS

CMP, DRY ETCHING, ION IMPLANTATION

METROLOGY

AFM

INTERFACE TRAPS FOR Q_s



WHY

ELECTRONS ARE CAPTURED BY UNPAIRED ELECTRON. UNPAIRED ELECTRON STRUCTURAL MISMATCH BETWEEN SI/SIO2

RELATED PROCESS

THERMAL OXIDATION

THRESHOLD VOLTAGE

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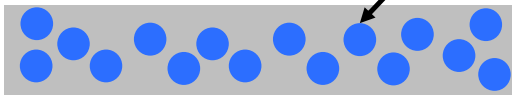
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<WHAT FACTORS AFFECT THRESHOLD VOLTAGE?>

ACCEPTOR DOPING CONCENTRATION FOR N_A

Acceptor



WHY

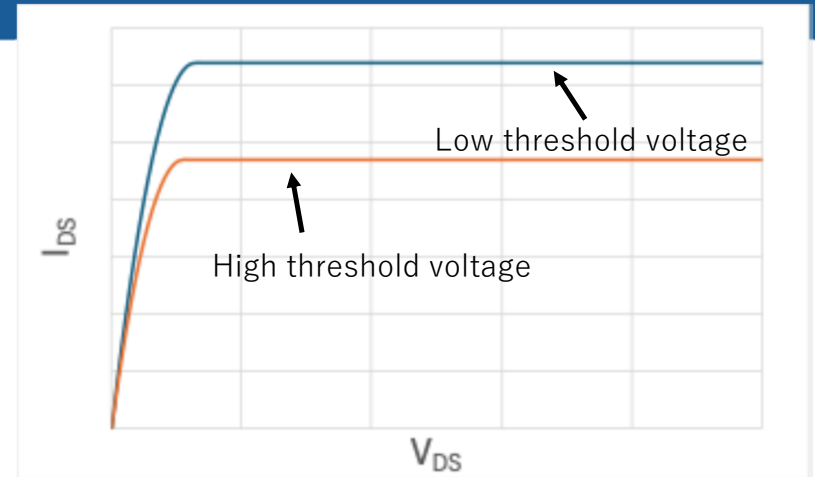
A HIGHER ACCEPTOR DOPING CONCENTRATION MAKES IT HARDER TO FORM THE INVERSION LAYER.

RELATED
PROCESS

WELL FORMATION, SUBSTRATE PREPARATION

METROLOGY

SIMS



AS THE THRESHOLD VOLTAGE INCREASES, THE PINCH-OFF VOLTAGE DECREASES, AND THE SATURATION CURRENT ALSO BECOMES SMALLER.

THRESHOLD VOLTAGE

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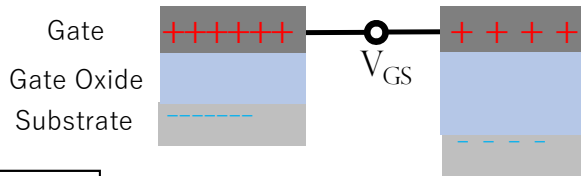
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<WHAT FACTORS AFFECT THRESHOLD VOLTAGE?>

GATE OXIDE THICKNESS FOR C_{OX}

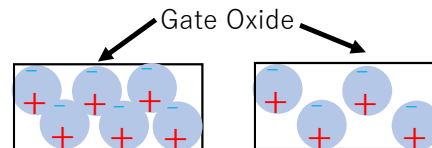


WHY THINNER OXIDE CAN INDUCE A LARGER AMOUNT OF CHARGE IN THE CHANNEL WITH A SMALLER GATE VOLTAGE.

RELATED PROCESS THERMAL OXIDATION

METROLOGY ELLIPSOMETRY, XRR

FILM DENSITY FOR C_{OX} , ϵ_s



WHY LOWER FILM DENSITY MEANS FEWER ATOMS TO POLARIZE, SO THE DIELECTRIC CONSTANT GETS SMALLER

RELATED PROCESS THERMAL OXIDATION

METROLOGY ELLIPSOMETRY (MEASURE REFLECTIVITY)

TYPE OF GATE OXIDES FOR C_{OX} , ϵ_s

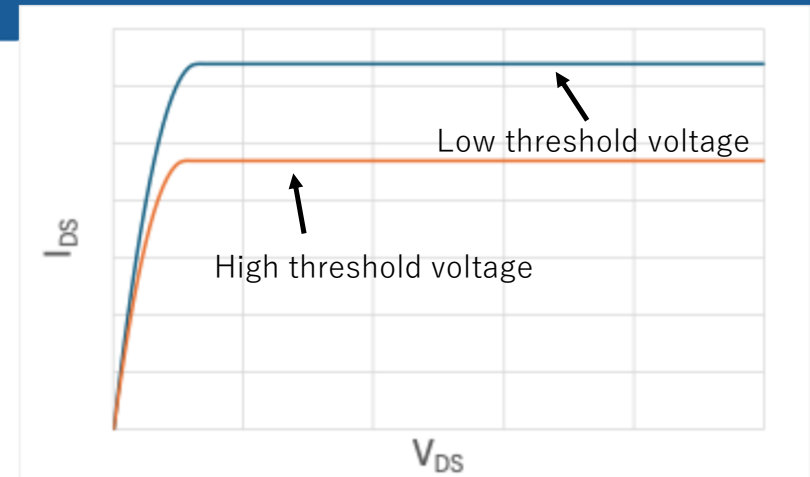
SiO_2
 $\epsilon = 3.9$

Si_3N_4
 $\epsilon = 7$

HfO_2
 $\epsilon = 20 \sim 25$

WHY DIELECTRIC CONSTANT VARIES WITH POLARIZABILITY OF THE MATERIAL

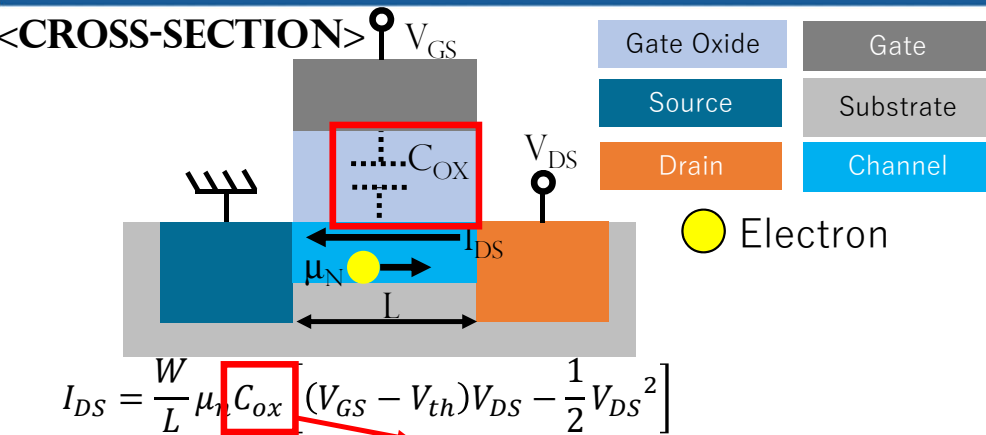
RELATED PROCESS THERMAL OXIDATION, ALD, CVD



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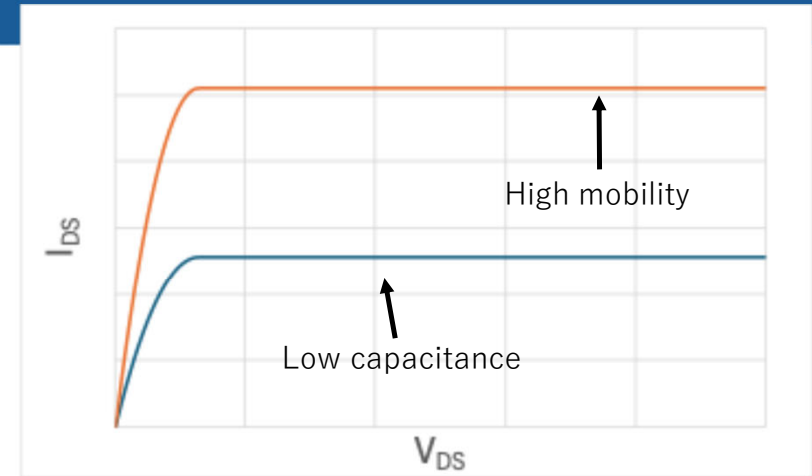
GATE CAPACITANCE

6



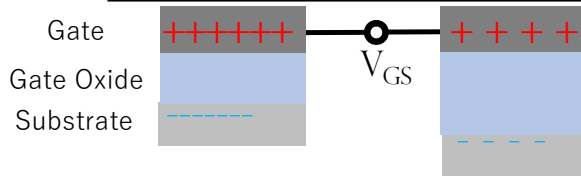
$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} \rightarrow \begin{matrix} \text{PERMITTIVITY} \\ \text{THICKNESS} \end{matrix}$$

GATE OXIDE CAPACITANCE IS PROPORTIONAL TO DRAIN-SOURCE CURRENT
IF CAPACITANCE DECREASES LOWER, DRAIN-SOURCE CURRENT ALSO DECREASES



<WHAT FACTORS AFFECT OXIDE CAPACITANCE?>

GATE OXIDE THICKNESS



WHY

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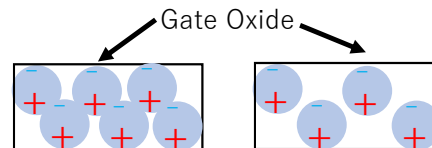
RELATED PROCESS

THERMAL OXIDATION

METROLOGY

ELLIPSOMETRY, XRR

FILM DENSITY



WHY

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THERMAL OXIDATION
ELLIPSOMETRY (MEASURE REFLECTIVITY)

RELATED PROCESS

METROLOGY

TYPE OF GATE OXIDES

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RELATED PROCESS

THERMAL OXIDATION, ALD, CVD